

Visual Instruction Tuning: A Review of LLaVA

MD Farhan¹ (2312703042), Sayeed Mortuza² (2311448042)

Supervisor: Mohammad Shifat-E-Rabbi

Department of Computer Science and Engineering

ABSTRACT

Abstract—This report summarizes and reviews *Visual Instruction Tuning*, the paper that introduced LLaVA (Large Language and Vision Assistant). LLaVA is the first system to use a language-only GPT-4 model to generate multimodal instruction-following data, which is then used to instruction-tune an end-to-end vision-language model built from a CLIP visual encoder and a Vicuna LLM. We describe the data-generation pipeline, model architecture, two-stage training procedure, and the reported results on the LLaVA-Bench and ScienceQA benchmarks, and close with a discussion of the paper’s contributions and limitations.

1 INTRODUCTION

Large language models (LLMs) such as GPT-4 and ChatGPT have shown that instruction tuning greatly improves an LLM’s ability to follow human intent in a zero-shot manner. This idea was largely unexplored in the multimodal (vision + language) setting prior to this work. The paper *Visual Instruction Tuning* (Liu *et al.*, NeurIPS 2023) addresses this gap by proposing a pipeline to generate visual instruction-following data using a text-only GPT-4, and by training LLaVA, an end-to-end multimodal model that connects a vision encoder to an LLM. The paper’s main contributions are: (1) a GPT-4-based data generation pipeline for multimodal instructions, (2) the LLaVA architecture and two-stage training recipe, (3) two new evaluation benchmarks (LLaVA-Bench), and (4) full open-sourcing of data, code, and model weights.

2 DATA GENERATION PIPELINE

Since large-scale human-annotated vision-language instruction data did not exist, the authors used symbolic text representations of images—namely *captions* and *bounding boxes*—to prompt a text-only GPT-4/ChatGPT to synthesize instruction-following samples for COCO images. Three response types were generated using in-context seed examples:

- **Conversation:** multi-turn Q&A about visible objects, counts, actions, and relative positions.
- **Detailed description:** a single rich, comprehensive description of the image.
- **Complex reasoning:** step-by-step reasoning questions that go beyond direct visual recognition.

This process yielded **158K** unique language-image instruction samples (58K conversation, 23K detailed description, 77K complex reasoning). GPT-4 was found to consistently produce higher-quality data than ChatGPT, especially for spatial reasoning.

3 MODEL ARCHITECTURE

LLaVA connects a frozen pretrained **CLIP ViT-L/14** visual encoder to the **Vicuna** LLM via a single trainable linear projection matrix $\mathbf{W}$, mapping visual features $\mathbf{Z}_v = g(\mathbf{X}_v)$ into the LLM’s word-embedding space: $\mathbf{H}_v = \mathbf{W} \cdot \mathbf{Z}_v$. This lightweight scheme was chosen over more complex fusion mechanisms (e.g., Flamingo’s gated cross-attention, BLIP-2’s Q-Former) to allow rapid, data-centric iteration.

4 TRAINING PROCEDURE

Training proceeds in two stages:

1. **Feature alignment pre-training:** on a filtered 595K-pair subset of CC3M, only the projection matrix $\mathbf{W}$ is trained (both CLIP and the LLM stay frozen), aligning image features with the LLM’s word embeddings.
2. **End-to-end fine-tuning:** on the 158K instruction dataset (chatbot use case) or on ScienceQA, both $\mathbf{W}$ and the LLM weights are updated while the visual encoder remains frozen.

5 EXPERIMENTAL RESULTS

5.1 Multimodal Chatbot

On qualitative examples (e.g., the “extreme ironing” GPT-4 image), LLaVA produced responses that closely matched GPT-4’s reasoning quality, clearly outperforming BLIP-2 and OpenFlamingo, which tended to merely describe the image rather than follow the instruction. On the proposed **LLaVA-Bench (COCO)**, using all three data types achieved the best relative score of **85.1%** against a text-only GPT-4 reference; ablations showed each data type (especially complex reasoning) contributes meaningfully. On **LLaVA-Bench (In-the-Wild)**, LLaVA scored **67.3%** overall, outperforming BLIP-2 (+29%) and OpenFlamingo (+48%), though it still showed weaknesses in fine-grained perception and hallucination on out-of-domain, knowledge-heavy questions.

5.2 ScienceQA

Fine-tuned on ScienceQA, LLaVA alone achieved **90.92%** accuracy, close to the prior state of the art (91.68%). Combining LLaVA with a text-only GPT-4 as a judge/ensembler yielded a new state-of-the-art accuracy of **92.53%**, demonstrating that GPT-4 can be used effectively for model ensembling even without direct image access.

6 DISCUSSION AND LIMITATIONS

The paper shows that instruction tuning transfers effectively to the multimodal domain even with a comparatively small ($\sim 80\text{K}$ unique images), fully synthetic dataset. Reported limitations include hallucinated details not grounded in the image, sensitivity to CLIP’s fixed-resolution encoding (limiting fine detail/OCR-like perception), reliance on symbolic (text-only) image representations during data generation, and GPT-4-judged evaluation, which may introduce its own biases. The simple linear projection, while efficient, is a potential fusion bottleneck.

7 CONCLUSION

Visual Instruction Tuning established a practical, scalable recipe for building instruction-following multimodal assistants by distilling GPT-4’s language capabilities into visual instruction data. LLaVA achieved strong open-source chatbot performance and a new SoTA on ScienceQA, and its data pipeline and benchmarks (LLaVA-Bench) remain widely used foundations for later multimodal LLM research.

Source: H. Liu, C. Li, Q. Wu, Y. J. Lee, “Visual Instruction Tuning,” NeurIPS 2023, arXiv:2304.08485.